DRIVEABILITY—ENGINE RUNS ROUGH OR HESITATES WITH POSSIBLE MIS-FIRE DIAGNOSTIC TROUBLE CODES (DTC) P0301-P0308—WATER CONTAMINATION IN SPARK PLUG WELL—3.9L ENGINE ONLY	Article No. 02-16-2

FORD: 2002 THUNDERBIRD

LINCOLN: 2000-2002 LS

ISSUE

Some vehicles may exhibit an engine misfire or rough running condition. The Malfunction Indicator Lamp (MIL) may be illuminated with one or more mis-fire Diagnostic Trouble Codes (DTCs) stored. This could be caused by water entering the spark plug well area of the engine. This may occur after the front of the vehicle or engine compartment is exposed to an abnormal amount of water. During this condition, water could pass through the protective convolute on the engine wire harness and eventually drain into the spark plug wells.

ACTION

Remove a small section of convolute from the effected Ignition Coil wire harness allowing any water that may accumulate to drain away from the spark plug wells in the engine. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

1. Perform the following procedure only on the bank of the engine where water is present in the spark plug wells.
   - DTCs P0301-4 are the Right (passenger) side of the engine
   - DTCs P0305-8 are the Left (driver) side of the engine
2. Depending on the DTCs observed, remove the corresponding ignition coil cover from the engine and inspect for water intrusion. Refer to Workshop Manual, Section 303-07, (303-07B on LS).

NOTE

IF WATER IS NOT PRESENT IN THE SPARK PLUG WELLS OF THE ENGINE, DO NOT PROCEED WITH THIS ARTICLE. CONTINUE WITH NORMAL POWERTRAIN CONTROL/ EMISSION DIAGNOSIS MANUAL DIAGNOSTIC PROCEDURES.

3. If water is present on the Left bank of the engine, proceed directly to Step 4. If water is present on the RIGHT bank of the engine, continue with the following steps:
   a. Remove the cabin air filter. Refer to Workshop Manual, Section 412-01.
   b. Remove cabin air filter housing and fresh air inlet duct.
   c. Remove the cross-vehicle support bar secured to the strut towers.
   d. Use a wet/dry vacuum or shop air to remove all water and debris from the area. This is an effort to prevent water from entering the engine when the effected spark plug(s) are removed.
   e. Disconnect all 4 (four) wire harness connectors from the ignition coils.
   f. Unsecure the ignition coil wiring harness retainer from the rear of the cam cover. Refer to Workshop Manual, Section 303-01, (303-01B on LS).
   g. Remove the effected ignition coil and spark plug and inspect. Replace only if damage from corrosion is present. Ignition coil part number 2W4Z-12029-BA, and spark plug part number AGSP-32F-SMF4.
   h. Install spark plug(s). Torque to 19 Lb-ft (26 N•m).
   i. Install ignition coil(s). Torque fastener to 44 Lb-in (5 N•m).
   j. Locate the end of the convolute protective cover on the wire harness near where it passes under the ignition coil cover. Carefully remove the foam insulation and peel back any tape in the area to expose approximately 2 inches of convolute.

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k. Remove the 2 inches of exposed convolute material covering the wire harness. This will allow any water that accumulates in the wire harness to drain away from the ignition coil cover. Use extreme caution not to damage the wire harness (Figure 1).

l. Remove all dirt and moisture from the notch in the cam cover and ignition coil cover where the harness passes through.

m. Route the ignition coil wire harness into the cam cover and connect the ignition coil wire connectors.

n. Lift the wire harness up and apply a 1/2 inch bead of silicone sealer, part number F7AZ-19554-EA or equivalent, into the notch in the cam cover (Figure 2).

o. Set the ignition coil wire harness in the bead of silicone sealer and apply enough additional silicone sealer to surround the wire harness and notch (Figure 3).

p. Install the ignition coil cover. Torque the bolts to 44 lb-in (5 N•m).

NOTE

THERE MUST BE ENOUGH CONVOLUTE REMOVED SO THAT APPROXIMATELY 1 INCH OF EXPOSED WIRE HARNESS IS VISIBLE OUTSIDE THE REAR OF THE IGNITION COIL COVER WHEN ASSEMBLED (FIGURE 4).

q. Secure the ignition coil wire harness to the retainer in the rear of the cam cover.

r. Install the fresh air inlet duct. Torque fasteners to 53 Lb-in (6 N•m).

s. Install the cabin air filter housing and filter. Torque fasteners to 53 Lb-in (6 N•m). Refer to Workshop Manual Section 412-01.

t. Install the cross-vehicle support bar. Torque the fasteners to 15 Lb-ft (20 N•m).

u. Clear DTCs and verify the concern has been repaired.

4. If water is present on the LEFT bank of the engine continue with the following steps:

a. Remove the cowl vent screen. Refer to Workshop Manual, Section 501-02.

b. Remove the cross-vehicle support bar secured to the strut towers.

c. Remove the evaporative canister purge valve cover.

d. Use a wet/dry vacuum or shop air to remove all water and debris from the area. This is an effort to prevent water from entering the engine when the effected spark plug(s) are removed.

e. Disconnect all 4 (four) wire harness connectors from the ignition coils.

f. Unsecure the ignition coil wiring harness retainer from the rear of the cam cover. Refer to Workshop Manual, Section 303-01, (303-01B on LS).

g. Remove the effected ignition coil and spark plug and inspect. Replace only if damage from corrosion is present. Ignition coil part number 2W4Z-12029-BA, and spark plug part number AGSP-32F-SMF4.

h. Install spark plug(s). Torque to 19 Lb-ft (26 N•m).

i. Install ignition coil(s). Torque fastener to 44 Lb-in (5 N•m).

j. Locate the end of the convolute protective cover on the wire harness where it passes under the ignition coil cover. Carefully remove the foam insulation and peel back any tape in the area to expose approximately 2 inches of convolute.

k. Remove the 2 inches of exposed convolute material covering the wire harness. This will allow any water that accumulates in the wire harness to drain away from the ignition coil cover. Use extreme caution not to damage the wire harness (Figure 1).

l. Remove all dirt and moisture from the notch in the cam cover and ignition coil cover where the harness passes through.

m. Route the ignition coil wire harness into the cam cover and connect the ignition coil wire connectors.

n. Lift the wire harness up and apply a 1/2 inch bead of silicone sealer, part number F7AZ-19554-EA or equivalent, into the notch in the cam cover (Figure 2).

o. Set the ignition coil wire harness in the bead of silicone sealer and apply enough additional silicone sealer to surround the wire harness and notch (Figure 3).

p. Install the ignition coil cover. Torque the bolts to 44 Lb-in (5 N•m).

NOTE
THERE MUST BE ENOUGH CONVOLUTE REMOVED SO THAT APPROXIMATELY 1 INCH OF EXPOSED WIRE HARNESS IS VISIBLE OUTSIDE THE REAR OF THE IGNITION COIL COVER (FIGURE 4).

q. Secure the ignition coil wire harness to the retainer in the rear of the cam cover.

r. Install the cross-vehicle support bar. Torque the fasteners to 15 Lb-ft (20 N•m).

s. Install the cowl vent screen. Refer to Workshop Manual, Section 501-02.

t. Install the evaporative canister purge valve cover.

u. Clear DTCs and verify the concern has been repaired.

PART NUMBER	PART NAME
2W4Z-12029-BA	Ignition Coil
AGSP-32F-SMF4	Spark Plug
F7AZ-19554-EA	Adhesive & Sealer

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And Emissions Warranty Coverage

OPERATION	DESCRIPTION	TIME
021602A	Seal Ignition Coil Wires At Right Hand Coil Cover	1.5 Hrs.
021602B	Seal Ignition Coil Wires At Left Hand Coil Cover	1.4 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
12B637	95

OASIS CODES: 608000, 608400, 608500, 608600, 609000, 609400, 609500, 609600, 698298

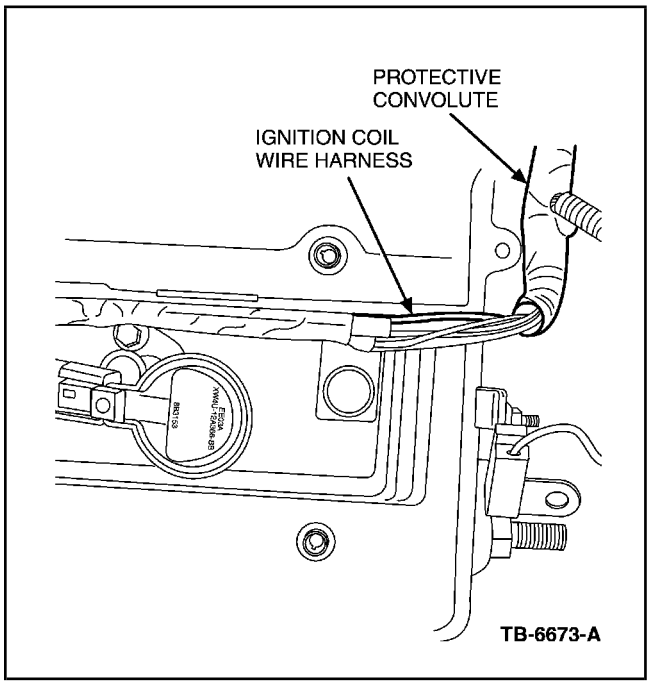


Figure 1 - Article 02-16-2

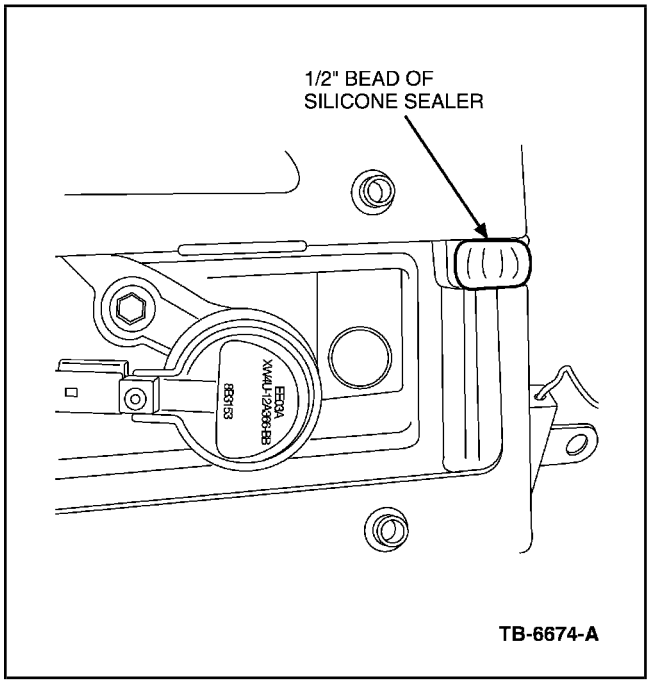


Figure 2 - Article 02-16-2

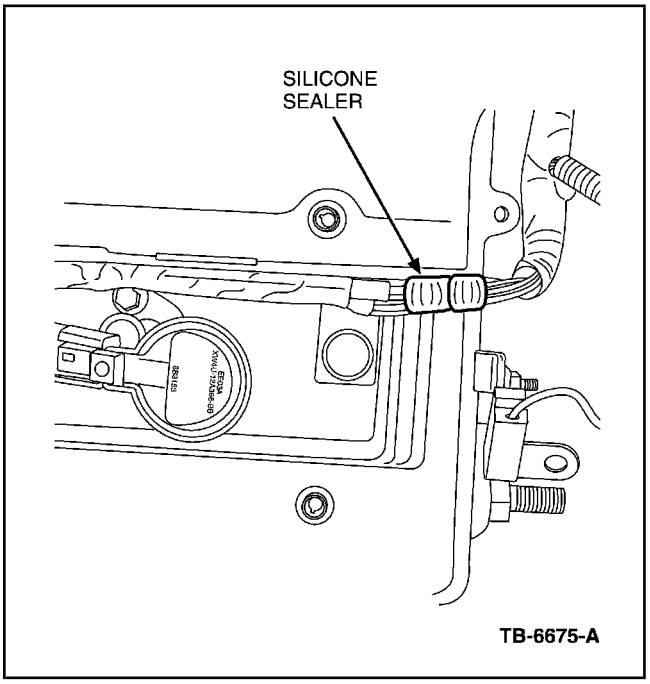


Figure 3 - Article 02-16-2

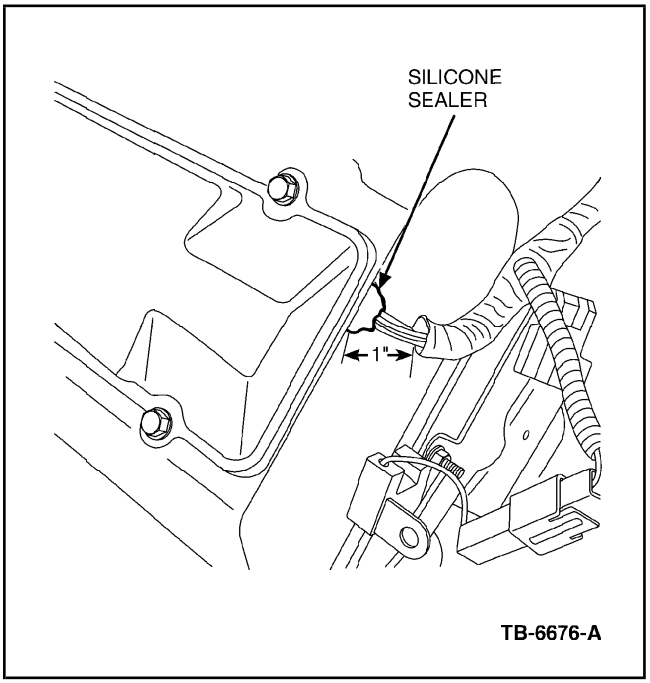


Figure 4 - Article 02-16-2